



Case Report

Invasive Aspergillosis with impaired neutrophil responses against *Aspergillus fumigatus* in patients treated with Acalabrutinib—findings from three cases



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ABSTRACT

Objectives: Ibrutinib, a first-generation covalent Bruton's tyrosine kinase inhibitor (BTKi) was found to be a risk factor for the occurrence of invasive fungal complications. Acalabrutinib is a second-generation covalent BTKi used to treat B-cell malignancies. Healthy donor neutrophils incubated *ex vivo* with acalabrutinib lose ability to control *Aspergillus* conidia germination. In patients receiving acalabrutinib, the potential effect on neutrophil antifungal activity is unknown. Furthermore, only two cases of invasive aspergillosis have been reported during treatment with acalabrutinib, outside of a few cases in a clinical trial.

Methods: We describe three new cases of invasive aspergillosis occurring within the first months of acalabrutinib therapy in patients with chronic lymphocytic leukemia. We used videomicroscopy and flow cytometry approaches to investigate the basic functional responses against *Aspergillus* of neutrophils from acalabrutinib-treated patients.

Results: We showed an alteration in the anti-*Aspergillus* response after 1 month of acalabrutinib therapy: neutrophils lost their capacities of killing *Aspergillus fumigatus* germinating conidia and decreased their reactive oxygen species production when stimulated by *Aspergillus*.

Conclusions: It is important to follow-up patients treated with acalabrutinib for the risk of aspergillosis as well as those treated with ibrutinib.

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Introduction

B-cell malignancies such as chronic lymphocytic leukemia (CLL) or Waldenstrom's macroglobulinemia are now commonly treated

with inhibitors of Bruton's tyrosine kinase (BTK). Ibrutinib is a first-generation covalent BTK inhibitor (BTKi) and that was the first of its kind to be approved by the Food and Drug Administration. Ibrutinib was found to be a risk factor for the occurrence of invasive fungal complications, mainly invasive aspergillosis, of which up to 40% were complicated by cerebral involvement [1].

We and others have shown defects in the anti-*Aspergillus* immune response in association with ibrutinib administration [2,3].

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We identified deficits affecting the neutrophil, a key cell in the defense against *Aspergillus* [4]. The main defects were reduced production of reactive oxygen species (ROS) and loss of the ability to engulf and kill *Aspergillus* hyphae.

Acalabrutinib, a second-generation covalent BTKi in use since 2017 is expected to have fewer side effects given its enhanced selectivity for BTK [5]. An altered anti-*Aspergillus* response has been observed in healthy donor cells incubated with acalabrutinib [3], but in patients receiving acalabrutinib, the potential effect on neutrophil antifungal activity is unknown. Furthermore, only two cases of invasive aspergillosis have been reported during treatment with acalabrutinib [6,7], apart from a few poorly detailed cases in a clinical trial [8]. Herein, we describe three cases of invasive aspergillosis in CLL patients treated with acalabrutinib and investigate the basic functional responses against *Aspergillus* of neutrophils from acalabrutinib-treated patients.

Materials and methods

Case reports

Cases were reported from three hospitals in France between January and December 2022 (Supplementary Table 1). The total number of patients treated with acalabrutinib was available for two of the three hospitals.

Antifungal response study

Blood samples were collected from patients (distinct from reported cases) without BTKi and after 1 month of acalabrutinib, excluding patients receiving corticotherapy. Neutrophils were subjected to functional analysis as previously described [2].

Killing of *Aspergillus*

Neutrophils isolated from whole blood were incubated with fluorescent DsRed *A. fumigatus* germinating conidia (GC) (ratio: 16neutrophils: 1GC) in a 96-well plate at 37°C overnight. Sytox Green dye (1µM) was added to the culture medium to visualize *Aspergillus* death. Videos were acquired using a Zeiss Axio-Z1 fluorescence microscope (Carl Zeiss) and Zen software was used to quantify *Aspergillus* killing by neutrophils.

ROS production

Whole blood was incubated with Rhodamin-123 dye (10µM) to assess ROS production and then stimulated with *Aspergillus fumigatus* GC for 2 hours at 37°C. Flow cytometry was performed on a Fortessa-LSR (BD); gating neutrophils on FSC/SSC and cluster of differentiation CD15+/CD16+.

Case 1

An 80-year-old man, diagnosed with CLL for 16 years, experienced CLL progression. Acalabrutinib was initiated (day 1 [D1]) after 1 month of corticosteroid therapy (>0.3 mg/kg/d) for autoimmune hemolytic anemia. A chest computed tomography (CT) scan on D14 revealed multiple lung nodules. At D21, acalabrutinib was discontinued because of deterioration in general condition and neurological symptoms. *Aspergillus fumigatus* polymerase chain reaction (PCR) (in-house), beta-glucan (Fungitell, Associates of Cape Cod), and galactomannan (Platellia, Bio-Rad) were positive in two successive blood samples (D28 and D33). Voriconazole was started on D29. Cerebrospinal fluid analysis (D30) showed negative results for galactomannan and *A.fumigatus* PCR. A brain lesion (temporal lesion with annular contrast) was observed on a scan on D35. The

lesions increased in number and size. The patient died on D61 because of probable invasive aspergillosis with cerebral involvement.

Case 2

A 68-year-old woman, with a 5-year history of CLL received first-line treatment with the combination of acalabrutinib and anti-CD20 monoclonal antibody (obinutuzumab) (D1). Corticosteroid therapy (1 mg/kg/d) was prescribed for inflammatory ocular neuropathy. On D14, she developed a fever. On D31, she was admitted to the intensive care unit for worsening respiratory distress. Acalabrutinib was stopped on D33. Three positive broncho-alveolar lavages (BALs) for *A.fumigatus* PCR (D33, D38, and D41) and *Pneumocystis jirovecii* PCR and hyphae observed on BAL D33, led to the diagnosis of probable invasive aspergillosis and pneumocystosis. *A.fumigatus* PCR was positive in one blood sample (D38). Trimethoprim/sulfamethoxazole was started on D33 and voriconazole on D36 but the patient died on D48.

Case 3

A 77-year-old man, with a 10-year history of CLL on hematologic surveillance only, received obinutuzumab because of bone marrow infiltration. Acalabrutinib was started a few weeks later because of disease progression (D1). On D48, a chest CT scan revealed two pulmonary nodules. Sputum (D50) was positive for *A. fumigatus* by culture and PCR. Lung biopsy (D63) confirmed proven invasive aspergillosis with positive *A.fumigatus* culture. Isavuconazole was started on D65. Acalabrutinib was discontinued on D66, but the patient died on D71.

Antifungal response study results

Live video microscopy of neutrophils co-cultured with *Aspergillus* GC showed a striking loss of antifungal activity after acalabrutinib therapy. The mean GC killing rate dropped from 38.1% (+/- 6%) in untreated patients (n = 3) to 10.2% (±5%) in patients after 1 month of acalabrutinib (n = 3) (Figure 1a-b). About ROS production, neutrophils from three patients before acalabrutinib therapy showed a 4-fold increase when challenged with *Aspergillus* GC whereas neutrophils from the same patients after 1 month of acalabrutinib therapy showed only a 1.8-fold increase (Figure 1c-d).

Discussion

Although acalabrutinib is a BTKi that is more specific to its target [5] than ibrutinib, the incidence of fungal complications during treatment appears to be identical. A phase II acalabrutinib safety study reported a 1.1% rate of fungal infections (either because of *Candida* sp, *Aspergillus* sp, *Cryptococcus* sp, or *Pneumocystis jirovecii*) [9]. In a phase III study comparing acalabrutinib and ibrutinib, reported fungal complications were 3.8% and 1.9%, respectively [8]. Beyond clinical trials, only two cases of invasive aspergillosis have been reported in patients treated with acalabrutinib [6,7].

Here we describe three cases of invasive aspergillosis in patients receiving acalabrutinib, one with cerebral involvement and another associated with *Pneumocystis* pneumonia. In cases 1 and 2, corticosteroid therapy >0.3mg/kg/day was administered in the month before starting acalabrutinib. We recognized that corticoid administration with this posology is a common risk factor for invasive aspergillosis, but the cerebral involvement (case 1) appears to be distinct and potentially linked to BTKi use [1]. The *Aspergillus/Pneumocystis* co-infection (case 2) is also uncommon. The clinical presentation of cerebral aspergillosis in BTKi-treated patients raises unresolved questions. In case 3, the only host criterion is the BTKi administration [10].

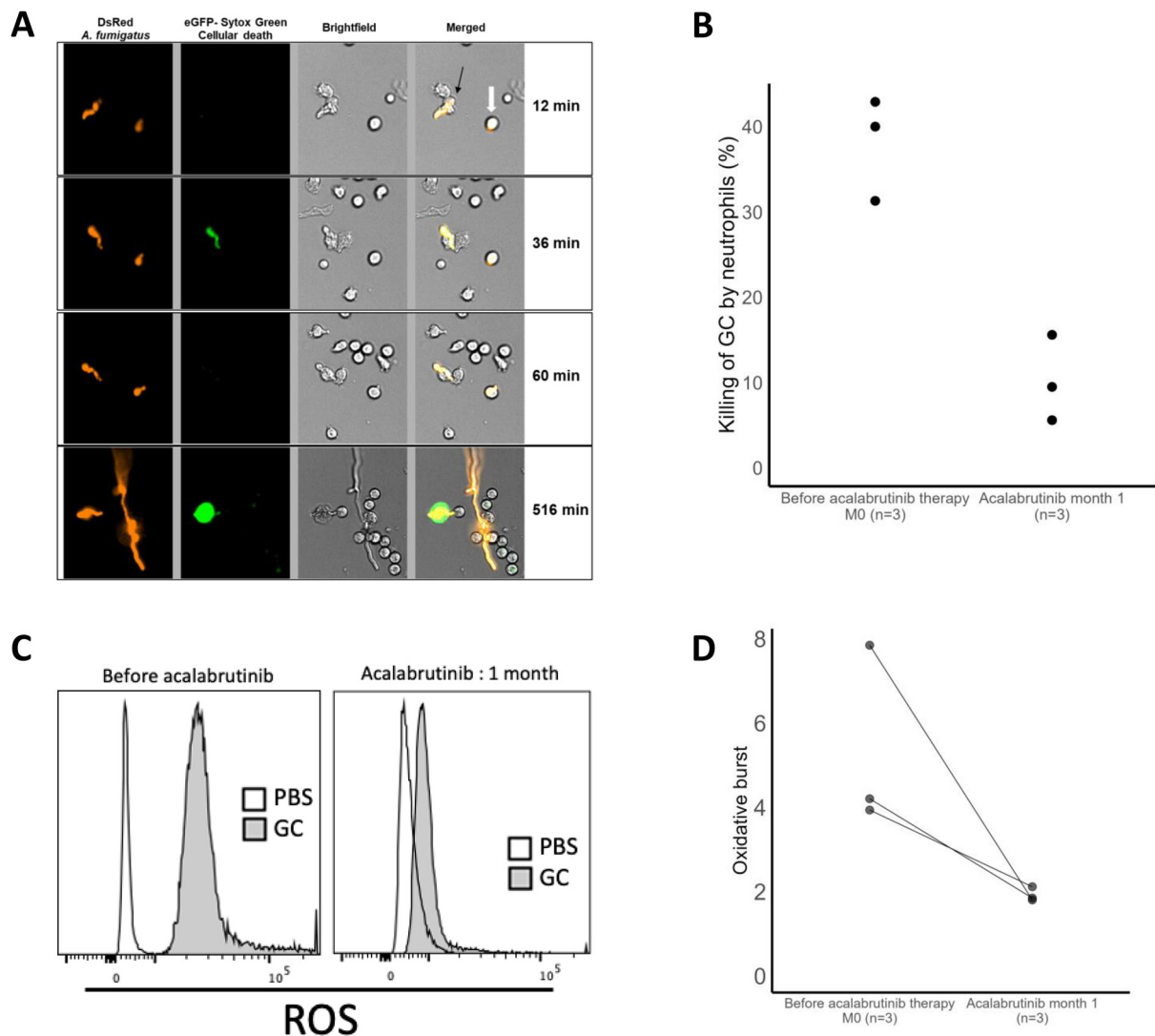


Figure 1. Neutrophils responses against *Aspergillus fumigatus* are impaired in patients after 1 month of acalabrutinib therapy. (a and b) Video-microscopy: Neutrophils are cultured with DsRed *Aspergillus fumigatus* germinating conidia (MOI 16:1) during 16 hours in fluorescence microscope. (a) Details of neutrophil-fungus interactions (neutrophils from healthy donor) observed at different co-culture times: **engulfment** of conidia by neutrophil (deformation and tight attachment of neutrophils around conidia ($t = 12$ min)), **killing** (thin black arrow: as indicated by green staining ($t = 36$ min)) during a short time of conidia with Sytox Green [eGFP] and no subsequent growth [$t = 516$ min]) and **killing failure** (white thick arrow: no staining with Sytox Green and growth [$t = 516$ min]). (b) Rate of killing of *A. fumigatus* germinating conidia by neutrophils from patients before (M0) ($n = 3$) and after 1 month ($n = 3$) of acalabrutinib treatment. (c and d) Whole blood was incubated for 2 hours with *A. fumigatus* germinating conidia or PBS as control before flow cytometry analysis. Rhodamin 123 dye was used to monitor ROS production. (c) ROS production of whole-blood neutrophils from a 54-year-old man with chronic lymphoid leukemia in control condition (white) or after stimulation with *A. fumigatus* germinating conidia (grey) before (left histogram) and after 35 days of acalabrutinib therapy (right histogram). (d) Oxidative burst of neutrophils from patients before (M0) and after 1 month (M1) of acalabrutinib therapy. Oxidative burst is defined for each point as the ratio of the mean fluorescence intensity of neutrophils ROS after stimulation by *A. fumigatus* germinating conidia to the mean fluorescence intensity in the basal state. eGFP, enhanced green fluorescent protein; MOI, multiplicity of infection; PBS, phosphate buffered saline; ROS, reactive oxygen species.

Similar to our previous results investigating the effects of ibrutinib, we observed compromised antifungal neutrophils activities (oxidative burst and killing capacities) after 1 month of acalabrutinib therapy.

These results are consistent with Colado et al. [3], who described diminished oxidative burst and a decrease in the ability of neutrophils to control *Aspergillus* conidia germination when healthy donors' neutrophils were incubated with acalabrutinib. Our work has the advantage of having been done in acalabrutinib-treated patients and thus of being able to assess the intra-patient evolution in neutrophil functions.

The scarcity of fungal complications, particularly invasive aspergillosis in patients with X-linked agammaglobulinemia who lack BTK suggests that the deleterious effect of BTKis on the anti-*Aspergillus* response is because of an off-target effect. Despite bet-

ter specificity against BTK, acalabrutinib also appears to have a negative impact on the anti-*Aspergillus* neutrophil response and the incidence of invasive aspergillosis in patients receiving acalabrutinib should be investigated. The available data from our series indicate an incidence of 4.3% (2/47), which must be treated with caution because of the limited number of patients.

Conclusion

We report three cases of invasive aspergillosis occurring within the first months of acalabrutinib therapy in patients with CLL. Functional analysis of neutrophils shows an alteration in the anti-*Aspergillus* response after acalabrutinib treatment. In the meantime, it seems important to follow up on patients treated with acalabrutinib for the risk of aspergillosis and those treated with

ibrutinib and to encourage all relevant teams to actively report any cases that occur.

Declarations of Competing Interest

The authors declare no conflict of interest in this study.

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Ethical approval

The study was conducted in accordance with the ethical standards of the institution and the declaration of Helsinki and was approved by local medical ethics committee (CPP Ile de France IV). A signed informed consent to participate in the study was obtained from patients.

Author contributions

MB designed the study, performed experiments, participated in data collection and analysis, performed statistical analysis and wrote the paper; GT performed experiments and participated in scientific discussions, data collection and analysis; AB participated in the design of the microscopy experiments, scientific discussions and drafting the paper; AP participated in inclusion of patients; FL participated in inclusion of patients and drafting the paper; CLPV participated in inclusion of patients; FS participated in inclusion of patients; MEB participated in inclusion of patients; JG participated in inclusion of patients and drafting the paper; AJ participated in inclusion of patients and drafting the paper; NS participated in inclusion of patients; AA participated in inclusion of patients; DRW participated in scientific discussions, inclusion of patients and drafting the paper; AF participated in the conception of the study, the experimental design and co-wrote the article.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ijid.2024.107000](https://doi.org/10.1016/j.ijid.2024.107000).

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